Distributed Coverage Control for Air–Ground Robot Systems With Heterogeneous Sensing Capabilities

Hang Zhang, Ronghao Zheng, *Member, IEEE*, Senlin Zhang, *Member, IEEE*, and Meiqin Liu, *Senior Member, IEEE*

***Abstract*—In multi-robot coverage control, ground robots aim to cover and monitor a domain optimally. However, when covering an extensive domain like a densely forested potential fire site, the sensing capabilities of ground robots are limited, resulting in poor coverage. Leveraging the aerial robots' ability to expand sensing ranges through high-altitude flight, this paper proposes a fully distributed, air-ground coverage control scheme to address this challenge. First, aerial robots provide a low-resolution coverage of the domain. Then, they use coarse but broad sensing information to guide ground robots, with short-range but high-resolution sensing, to achieve a high-resolution coverage. Simultaneously, each aerial robot dynamically adjusts its cell size to match its load, enhancing the coverage performance. The convergence of the control scheme is proved and its performance is evaluated through simulations and experiments.**

***Note to Practitioners*—This work is motivated by multi-robot coverage control, which is widely applied in environmental monitoring, search, disaster rescue, and positioning services. Existing coverage control schemes typically assume robots have long-range sensing capabilities. However, in practical scenarios such as covering large-scale dense forest fire sites, robots' sensing is limited. To address this, a fully distributed air-ground cooperative coverage control scheme is proposed. By expanding sensing ranges through high-altitude flights, aerial robots acquire coarse but extensive environmental data, which assists sensing-limited ground robots in achieving high-resolution coverage. The effectiveness of the scheme is rigorously proved through theoretical analysis and experimental validation involving multiple UAVs and UGVs.**

***Index Terms*—Coverage control, heterogeneous sensing, air-ground robot systems.**

I. Introduction

In RECENT years, there has been growing research interest in the coverage control of multi-robot systems, which has been widely applied in environmental monitoring [1], search [2], disaster rescue [3], and positioning service [4]. In these applications, robots are driven to seek optimal positions based on the local environmental observations with the goal of achieving optimal coverage needs.

The issue of coverage control [5], commonly tackled using the geometric tool known as Voronoi partition, is referred to as Voronoi-based coverage control [6], [7], [8]. In these works, robots partition the domain into Voronoi cells and employ a move-to-centroid controller to locate optimal positions for maximizing collective environmental sensing performance. While employing this controller, robots are often assumed to have long sensing ranges to be aware of the scalar field within their cells. However, when the domain to be covered is extensive, such as a dense forest, the ground robots' sensing is limited. Although these ground robots provide high-resolution sensing information, their limited sensing range prevents them from covering a large domain, leading to poor performance. To address this challenge, we utilize aerial robots with long-range but low-resolution sensing to assist, and propose a fully distributed control scheme for the air-ground robot systems.

A. Related Work on Sensing-Limited Robots' Coverage

When multiple ground robots try to cover a large domain, due to their short sensing ranges, coverage control schemes must advance beyond the move-to-centroid controller. Reference [9] describes the problem as a mixed centroid-area problem and proposes a gradient-based controller, while [10] presents a distributed deterministic annealing algorithm to control robots to the final positions by computing temperature gradients. In [11] and [12], a r-limited Voronoi partitioning law is introduced to determine the cells for sensing-limited robots to solve the challenge. On the other hand, [13], [14], [15], [16] proposes a developed r-limited Voronoi partitioning law integrated with a partition-based control scheme. The partitioning law allows robots to partition the domain using only the information from their limited range. Based on the law, [14] validates the effectiveness of the law through real experiments with drones in the desert, and [15], [16] introduces a controller that combines exploration and coverage strategies for robots to cover an unknown environment. In [17], a gradient-based scheme is developed for a team of ground robots equipped with sensors of equal sensing range to maximize the coverage area. Meanwhile, [18] proposes a distributed gradient-based coordination algorithm for multiple robots with different omnidirectional sensing radii.

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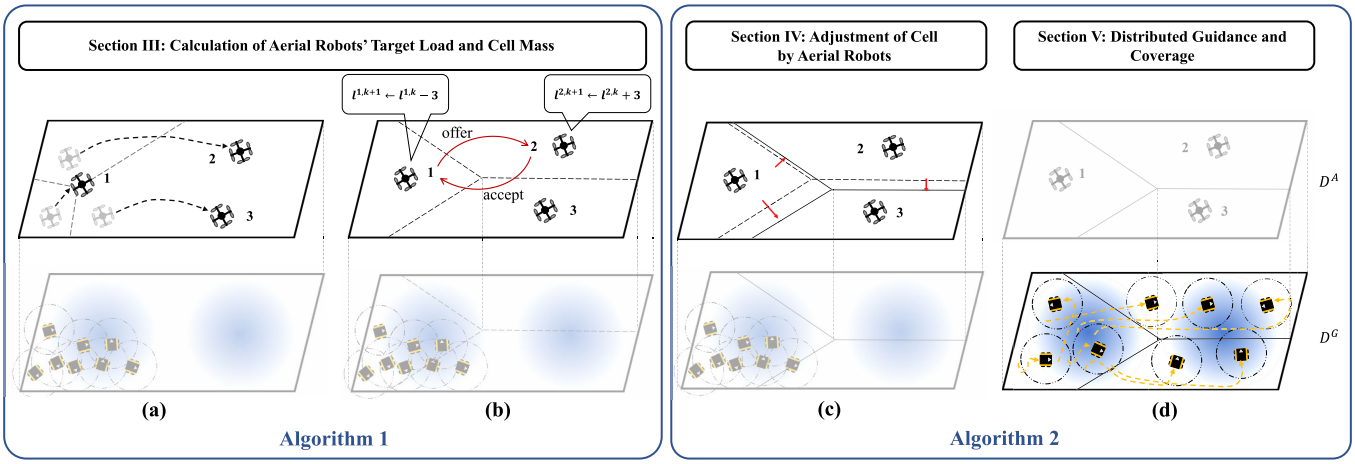


Fig. 1. The framework of this paper. Aerial robots operating in D^A utilize coarse but broader sensing information to assist ground robots in D^G in achieving high-resolution coverage.

In contrast, [19] introduces a continuous max-sum algorithm with gradient descent (CMGD), regardless of whether the sensing ranges are identical or different. Moreover, the coverage performance of CMGD is shown to be better than the methods presented in [17] and [18]. References [20] and [21] utilize weighted Voronoi cells to characterize the robots' sensing capabilities. They design controllers based on the varying sensing capabilities to achieve coverage tasks. The above extensions employ collaboration between sensing-limited robots to enhance coverage performance. However, as shown in Fig. 1, when the domain size is excessively vast, exceeding the collective sensing regions of all ground robots, coverage performance is poor.

B. Related Work on Air-Ground Robot Systems' Coverage

As highlighted in survey [22], hierarchical perception and control coordination across aerial and ground robots effectively accomplish tasks beyond the capability of individual robots. For instance, [23] uses aerial robots to construct maps for optimizing ground robot paths, and [24] decomposes the task into coarse exploration by ground robots and fine mapping by aerial robots. Therefore, to address the poor coverage performed by individual ground robots, some schemes also integrate the diverse sensing strengths of air-ground robot teams, providing better coverage for excessively vast domains. Aerial robots, with their ability to increase sensing range by flying at higher altitudes [25], [26], [27], are utilized to assist ground robots with limited sensing capabilities to achieve better coverage [28], [29], [30], [31], [32]. The aerial robots first achieve a rough coverage of the whole domain, then guide ground robots with their dominance to complete local but finer coverage. However, [28] requires a human operator to provide waypoints for the ground robot's safe navigation, while in [29], the guide controller relies on global information: the positions of all aerial robots, the total number of ground robots, and the total mass of the entire domain. Moreover, the fixed aerial robots' cells do not scale well with the number of ground robots inside, resulting in suboptimal coverage performance. Reference [30] proposes a distributed team-based scheme for coverage with heterogeneous robot teams. The weighted Voronoi diagrams are used to partition the domain among the teams and the ground robots within each team, based on their respective coverage capabilities. However, both the number of ground robots per team and the coverage capability of each team and each robot are pre-assigned.

C. Motivations and Contributions

Motivated by the scheme in [29], this paper proposes a fully distributed coverage control scheme for air-ground robot systems with heterogeneous sensing. As shown in Fig. 1, the aerial robots first achieve a low-resolution coverage, calculate the target load (the number of ground robots to supervise) and the matched cell mass (Algorithm 1). Then they adjust their cell size and guide ground robots simultaneously, achieving high-resolution coverage (Algorithm 2).

- In Algorithm 1, as shown in Fig. 1(a), the aerial robots first achieve a low-resolution coverage of the domain. Then, in Fig. 1(b), they will balance the supervised number of ground robots, ensuring nearly an equitable workload. Therefore, the loads of aerial robots 1 to 3 are determined as 3, 3, 2. Based on these loads, the target cell mass is also calculated to match the load to utilize the coverage capability of each ground robot fully.
- In Algorithm 2, given that the loads of aerial robots 1 and 2 are 3, which is more than that of aerial robot 3, the cell size of aerial robots 1 and 2 is adjusted to be larger to achieve the target cell mass, as depicted in Fig. 1(c). In Fig. 1(d), aerial robot 1 guides ground robots to the cell of aerial robot 2 and 3, thereby reaching the target loads.

The main contributions are fourfold:

1) Propose a distributed algorithm for aerial robots to calculate their target load, that is, the number of ground robots they will oversee, and the target cell size they are responsible for (Section III).
2) Propose distributed weight adjustment law (7) and control law (8) to adjust aerial robots' cell size jointly.

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Compared with [29], where cells are fixed, the proposed laws adjust cell size to match the number of ground robots, fully using the capability of each ground robot. Compared with [30], an additional control law (8) is designed to accelerate cell adjustment based on environment, not pre-assigned capability (Section IV).

3) Propose a fully distributed guidance and coverage scheme for ground robots to achieve high-resolution coverage. Compared with [29], aerial robots guide ground robots to achieve finer coverage in a distributed manner, without relying on global information (Section V).
4) To evaluate and compare the performance of the proposed scheme and multiple existing schemes, simulations and experiments are conducted in this paper, with the coverage performance of these schemes compared. The results show that the proposed scheme can achieve better coverage performance in the coverage control for ground mobile robots with sensing limitations (Section VI).

The rest of this paper is organized as follows. In Section II, we present preliminaries and formulate the coverage problem. From Sections III to V, we introduce the control scheme and thoroughly discuss the convergence. Section VI presents both simulation and experimental results. Section VII presents our conclusions and future work.

II. Problem Formulation

A. Preliminaries

Firstly, we recall the preliminaries of Voronoi-based coverage control [6]. Consider a convex $D \subset \mathbb{R}^2$ that is the domain of interest, for example, a potential fire site or a domain of chemical pollutant accumulation. The spatial distribution of temperature or pollutant concentration can be modeled as a scalar field, and the scalar field value of any point $q \in D$ is $\phi(q) : D \to \mathbb{R}_+$, which is bounded, integrable, and continuous with respect to q. Denote $G = \{1, 2, \ldots, N\}$. There are N ground robots with positions $p_i(t) \in D, i \in G$, trying to cover the domain D. Each ground robot i is given a Voronoi cell

$$V(p_i) = \left\{ q \in D \mid \|p_i - q\| \le \|p_j - q\|, j \in G \right\},$$

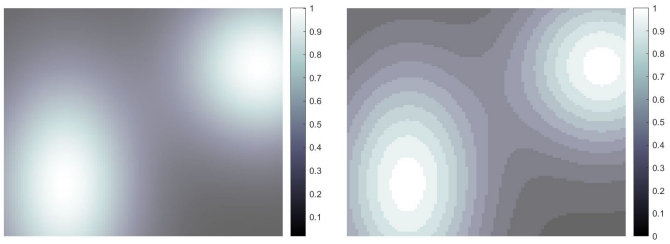
and the mass, the mass centroid of $V(p_i)$ is defined as

$$s(p_i) = \int_{V(p_i)} \phi(q)\, \mathrm{d}q, \quad C(p_i) = \frac{1}{s(p_i)} \int_{V(p_i)} q\phi(q)\, \mathrm{d}q.$$

To evaluate the performance of multi-robot coverage, a coverage cost function $\mathcal{H}_p$ is defined as follows, wherein a higher value of the function indicates a poorer coverage performance

$$\mathcal{H}_p(p_1, p_2, \ldots, p_N) = \sum_{i \in G} \int_{V(p_i)} \|q - p_i\|^2 \phi(q)\, \mathrm{d}q. \tag{1}$$

Therefore, the purpose of coverage control is to seek the corresponding optimal positions of ground robots, minimizing the above cost function (1).



(a) High resolution. (b) Low resolution.

Fig. 2. An example of high resolution and low resolution.

If the robots have large sensing ranges that encompass their cells, by taking the gradient of Eq. (1), [6] uses the move-to-centroid controller (2) to guide the robots to the optimal locations

$$\dot{p}_i = -\frac{\partial \mathcal{H}_p}{\partial p_i} = 2s(p_i)\left(C(p_i) - p_i\right). \tag{2}$$

B. Air-Ground Robot Team and Monitor Cell

To achieve better coverage in situations where ground robots' sensing is short-range, aerial robots are utilized as assistants. Consider a scenario where there are $K(> 1)$ aerial robots and $N(> K)$ ground robots in the domain. The ground robots operate within a domain $D^G \subset \mathbb{R}^2$, while the aerial robots operate above and parallel to D^G within a domain $D^A \subset \mathbb{R}^2$, as illustrated in Fig. 1. With short-range, high-resolution sensing, the ground robots can obtain high-resolution field values $\phi_{\mathrm{H}}(q)$ within their short-range sensing regions. Conversely, aerial robots, capable of flying higher to achieve broader but coarser sensing, can obtain low-resolution field values $\phi_{\mathrm{L}}(q)$ across a larger region. Fig. 2 illustrates an example of detecting the entire domain using high-resolution and low-resolution sensing. Here, "resolution" refers to the capacity to detect the minimum change in the scalar field value $\phi(q)$. In Fig. 2, the high-resolution means the ground robots can detect changes in the scalar field value $\phi(q)$ as small as 1×10^{-4}, while the low-resolution means the aerial robots can detect changes as small as 1×10^{-1} in the scalar field value $\phi(q)$.

There is a team of aerial robots $A = \{1, 2, \ldots, K\}$ with their positions denoted as $p^i(t) \in D^A$ for $i \in A$. Each aerial robot i is assigned a cell $V^i(t)$ defined as follows

$$\begin{aligned} V^i(t) = \big\{ & q \in D^A \mid \|p^i(t) - q\|^2 - w^i(t) \\ & \le \|p^j(t) - q\|^2 - w^j(t), j \in A \big\}, \end{aligned} \tag{3}$$

where $w^i(t)$ and $w^j(t)$ represent the weights associated with $V^i(t)$ and $V^j(t)$ respectively. And the sensing region of aerial robot i is defined by $B^i = \{q \in D^A \mid \|q - p^i(t)\| \le R^A\}$, where R^A is a sufficiently large sensing radius. Furthermore, we use an undirected graph $\mathcal{G} = (A, \mathcal{E})$ to describe the relationships between the aerial robots, where $\mathcal{E} \subseteq A \times A$ is the set of all edges. The set of aerial robot i's neighbors is denoted as $\mathcal{N}^i = \left\{ j \mid V^j(t) \cap V^i(t) \neq \emptyset, j \in A, j \neq i \right\}$.

Let $\mathcal{J}^i(t) = \left\{ j \mid p_j(t) \in V^i \right\}$ denote the set of ground robots under the dominance of aerial robot i, and $n^i(t) = |\mathcal{J}^i(t)|$

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representing the number of ground robots in cell V^i. Each ground robot j is also assigned a Voronoi cell

$$V_j(t) = \left\{q \in V^i \mid \|p_j(t) - q\| \le \|p_r(t) - q\|, r \in \mathcal{J}^i(t)\right\},$$

and its sensing region is limited by

$$B_j = \{q \in D^G \mid \|q - p_j(t)\| \le R^G\},$$

where R^G is the sensing radius. Furthermore, we denote the set of ground robot j's neighbors as $\mathcal{N}_j = \left\{r \mid V_r(t) \cap V_j(t) \neq \emptyset, r \in \mathcal{J}^i(t), r \neq j\right\}$, and make an assumption below regarding communication for the aerial and ground robots.

Assumption 1 (Communication Requirements): The communication range of each aerial and ground robot is sufficiently large. This range enables bidirectional communication between each aerial robot i with both its neighbors $\mathcal{N}^i$ and the ground robots $\mathcal{J}^i(t)$ within its cell $V^i(t)$. Additionally, each ground robot j can communicate with its neighbors $\mathcal{N}_j$.

The kinematic model of the aerial robot and the ground robot is described below

$$\frac{\mathrm{d}}{\mathrm{d}t}p^i(t) = u^i(t), \quad \frac{\mathrm{d}}{\mathrm{d}t}p_j(t) = u_j(t), \tag{4}$$

where $u^i(t)$ and $u_j(t)$ are the control inputs, for each $i \in A$ and $j \in G$. In the next, we omit the time variable t in time-dependent quantities only when this does not cause any confusion.

Remark 1: Using the adjust-communication-radius algorithm outlined in [6], the Voronoi cell $V^i(t)$ and $V_j(t)$ can be determined in a distributed manner by the neighbors $\mathcal{N}^i$ and $\mathcal{N}_j$ respectively.

Remark 2: Despite single integrator model's simplicity, it remains highly effective for real mobile robot control. With a trajectory tracking controller, it can generate velocity references for various platforms, including wheeled mobile robots [33] and autonomous aerial vehicles [34].

C. Problem Statement

Same as [6], [9], [13], and [29], the control objective is to minimize the coverage cost function. In the context of our problem description, the coverage cost function (1) can be equivalently written in the following form

$$\mathcal{H} = \sum_{j \in G} \int_{V_j} \|q - p_j(t)\|^2 \phi(q)\, \mathrm{d}q. \tag{5}$$

As illustrated in Fig. 1, this paper proposes a fully distributed scheme. In the scheme, Algorithm 1 and Algorithm 2 are sequentially executed to minimize the coverage cost function (5). Algorithm 1 will be introduced in Section III, while Algorithm 2 is detailed across Sections IV and V.

Remark 3: This work focuses on local minima of the coverage cost function $\mathcal{H}$, as global optimization is known to be NP-hard even in centralized settings [20], [35], [36]. In an attempt to find a better solution, methods such as the deterministic annealing algorithm [37] or incorporating an orthogonal exploration force [38] into the move-to-centroid controller can be employed.

Algorithm 1 Calculation of the Target Load and Cell Mass of Aerial Robot i

Driven by controller (6) to cover the domain D^G;
Create cell V^i (3) and obtain $n^{i,0}$ and $s^{i,0}$;
Set: $k \leftarrow 0, l^{i,0} \leftarrow n^{i,0}, y^{i,0} \leftarrow \frac{n^{i,0}}{s^{i,0}}$;
while $k < k_{\text{total}}$ **do**
  Acquire $l^{j,k}, y^{j,k}$ from its neighbor $j \in \mathcal{N}^i$ via communication;
  // **Providing Phase:**
  Select one neighbor j with minimum load $l^{j,k}$;
  if $l^{i,k} > l^{j,k}$ **then**
    Send the offer to aerial robot j;
  end
  // **Receiving Phase:**
  if receive offers **then**
    Select aerial robot r with maximum $l^{r,k}$;
    Send the acceptance to aerial robot r;
    Set: $l^{i,k+1} \leftarrow l^{i,k} + \lceil \frac{l^{r,k} - l^{i,k}}{2} \rceil$;
  end
  // **Transferring Phase:**
  if receive acceptances from neighbor h **then**
    Set: $l^{i,k+1} \leftarrow l^{i,k} - \lceil \frac{l^{i,k} - l^{h,k}}{2} \rceil$;
  end
  Set: $y^{i,k+1} \leftarrow y^{i,k} + \frac{g_y}{s^{i,0}} \sum_{j \in \mathcal{N}^i} (y^{j,k} - y^{i,k})$;
  Set: $k \leftarrow k + 1$;
end
Set: $\bar{n}^i \leftarrow l^{i,k_{\text{total}}}, \bar{s}^i \leftarrow \frac{\bar{n}^i}{y^{i,k_{\text{total}}}}$;

III. Calculation of Aerial Robots' Target Load and Cell Mass

In this section, Algorithm 1 is first presented for each aerial robot i to determine the number of ground robots and the cell size under its supervision. Then, we also discuss the computational complexity and the convergence of Algorithm 1.

First, the aerial robots are driven by the controller

$$u^i(t) = g_1 \left(C^i(t) - p^i(t)\right), \tag{6}$$

to provide a low-resolution coverage of the domain, where g_1 is the positive control gain and $C^i(t) = \int_{V^i(t)} q\, \mathrm{d}q / \int_{V^i(t)} \mathrm{d}q$ is the geometry centroid of V^i. When doing this, the aerial robots create cells in the form of weighted Voronoi cells V^i, where $w^i(t) = 0$ for all i. Then each aerial robot i has access to the local information: the number of ground robots $n^i(t)$ in its cell V^i, the mass $s^i(t)$ of cell V^i is defined as $s^i(t) = \int_{V^i(t)} \phi_{\mathrm{L}}(q)\, \mathrm{d}q$.

Under the controller (6), when the distance between $C^i(t)$ and $p^i(t)$ becomes very small, aerial robot i ceases its movement. When all the aerial robots cease their movement, a low-resolution coverage of the entire domain is completed. Subsequently, each aerial robot i obtains the number of the ground robots $n^{i,0}$ within its cell and the mass $s^{i,0}$ of its cell at that time. After that, aerial robot i calculates its target load $\bar{n}^i$ and matched cell mass $\bar{s}^i$ based on the initial value $n^{i,0}$ and $s^{i,0}$. The target load $\bar{n}^i$ is the average number of all ground robots, and the target cell mass $\bar{s}^i$ matches the target

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Algorithm 2 High-Resolution Coverage Achieved by Aerial Robot i and Ground Robots in the Cell V^i

repeat
  /* **Cell Adjustment:** */
  Update weight $w^i(t)$ (7) associated with $V^i(t)$;
  Driven by the controller (8);
  /* **Guidance and Coverage:** */
  // **Offering Phase:**
  Calculate $x^i(t)$ (9);
  Obtain $\underline{\mathcal{N}}^i(t)$ (10) and choose j_r from $\underline{\mathcal{N}}^i(t)$;
  if $x^i(t) > x^{j_r}(t)$ **then**
    Send offer to j_r consisting of $\left(i, x^i(t)\right)$;
  end
  // **Accepting Phase:**
  Check the received offer M_1 (11);
  if $|M_1| > 0$ **then**
    Obtain $\overline{\mathcal{N}}^i(t)$ (12) and choose h from $\overline{\mathcal{N}}^i(t)$;
    Send acceptance to h consisting of i;
  end
  // **Passing Phase:**
  Check the acceptance $M_2 = \{j\}$;
  if $|M_2| > 0$ **then**
    The ground robots in cell V^i are driven by the guide controller (13);
  else
    The ground robots in cell V^i are driven by the coverage controller (14);
  end
until a stopping condition is met (e.g., time t meets a certain requirement.);

load. When equal allocation of the total number of ground robots is not feasible, the maximum difference between the allocations is 1. In Algorithm 1, $l^{i,k}$ is the load value of aerial robot i at iteration k, g_y is a fixed positive gain constant, and k_{total} is a sufficiently large number of iterations because an individual robot cannot independently verify convergence using only neighbor information.

Motivated by [39], Lines 1 to 1 in Algorithm 1 is used to calculate target load value $\bar{n}^i$, which can be further divided into three main phases:

1) **Providing Phase:** Aerial robot i compares its current load value $l^{i,k}$ with those of its neighbors $\mathcal{N}^i$ and selects among its neighbors with minimum load $l^{j,k}$. If its load value satisfies that $l^{i,k} > l^{j,k}$, aerial robot i sends the offer to aerial robot j.
2) **Receiving Phase:** If aerial robot i receives the offers from its neighbors $\mathcal{N}^i$, it evaluates the offers from its neighbors $\mathcal{N}^i$. It selects aerial robot r with the highest load value $l^{r,k}$ and accepts its offer, then sends acceptance to aerial robot r and increases load value by $\lceil\left(l^{r,k} - l^{i,k}\right)/2\rceil$.
3) **Transferring Phase:** Aerial robot i checks if it has received acceptance. If it has received acceptance from aerial robot h, it reduces load value by $\lceil\left(l^{i,k} - l^{h,k}\right)/2\rceil$.

To calculate the cell mass $\bar{s}^i$ that matches $\bar{n}^i$, Line 1 is employed in Algorithm 1. By executing Line 1, $y^{i,k_{\text{total}}}$ is obtained. This value is then used to calculate the matched cell mass $\bar{s}^i$ via the division, i.e., Line 21.

For Algorithm 1, each aerial robot i is first driven by controller (6) and creates its cell V^i. Calculating the centroid C^i requires numerical integration over V^i, which costs $\mathcal{O}\left(M^i\right)$, where M^i is the number of discrete grid points in V^i. Creating the cell V^i involves communicating with neighbors, costing $\mathcal{O}\left(d^i\right)$, where d^i is the number of aerial robot i's neighbors. Additionally, computing the initial cell mass $s^{i,0}$ also requires integrating over V^i, costing $\mathcal{O}\left(M^i\right)$. After that, each aerial robot i iteratively exchanges load value $l^{i,k}$ and $y^{i,k}$ with its neighbors, requiring $\mathcal{O}\left(d^i\right)$ operations per iteration. This needs k_{total} iterations, so the total complexity for all K aerial robots is $\mathcal{O}\left(\sum_{i=1}^K M^i + \sum_{i=1}^K d^i + \sum_{i=1}^K M^i + k_{\text{total}} \cdot \sum_{i=1}^K d^i\right)$. This is equal to $\mathcal{O}\left(2M + \sum_{i=1}^K d^i + k_{\text{total}} \cdot \sum_{i=1}^K d^i\right)$, where M is the number of discrete grid points in the whole domain. Since the average number of Voronoi neighbors per aerial robot is bounded by a constant (at most six [40]), the computational complexity of Algorithm 1 can also be expressed as $\mathcal{O}\left(M + (k_{\text{total}} + 1) \cdot K\right)$.

Define the constant $S := \int_{D^G} \phi_{\text{L}}(q)\,\mathrm{d}q$, then Theorem 1 is presented, which guarantees the convergence of Algorithm 1.

Theorem 1: In the execution of Algorithm 1, if $k \to \infty$, $l^{i,k}$ will converge to $\lceil N/K \rceil$ or $\lfloor N/K \rfloor$, while satisfying that $\sum_{i=1}^K l^{i,k} = N$. And there exists a positive constant $g_y < \min_{i \in A}\left(s^{i,0}/|\mathcal{N}^i|\right)$ such that $y^{i,k}$ converges to N/S, implying that $\bar{n}^i/\bar{s}^i$ will be equal to N/S.

Proof: A proof is derived in Appendix A. ∎

Theorem 1 shows that if $k \to \infty$, $\bar{s}^i$ is equal to $\bar{n}^i S/N$. It indicates that the calculated target mass value $\bar{s}^i$ of each cell matches its target load value $\bar{n}^i$, which is important for fully utilizing the coverage capability of each ground robot.

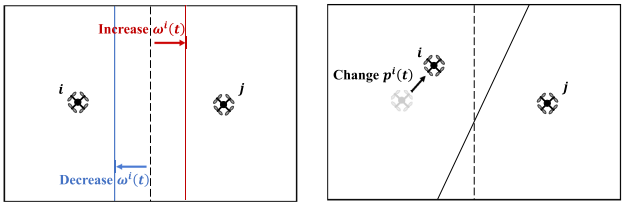
Remark 4: Unlike [29], which calculates aerial robots' target load and cell mass using global information (i.e., the total number of ground robots and the total mass of the domain), Algorithm 1 calculates these in a distributed manner.

IV. Adjustment of Cell by Aerial Robots

After each aerial robot i has obtained $\bar{n}^i$ and $\bar{s}^i$, it starts to adjust its cell size and guide ground robots to achieve high-resolution coverage at time t_0. This section introduces the adjustment of cell size by aerial robots, and the distributed guidance and coverage scheme will be discussed in Section V. Both processes are summarized in Algorithm 2.

The cell $V^i(t)$ (3) depends on the robot's position $p^i(t)$ and its associated weight $w^i(t)$. As illustrated in Fig. 3, varying either $p^i(t)$ or $w^i(t)$ alters the size and shape of $V^i(t)$. Specifically, adjusting the weight $w^i(t)$ shifts the cell boundaries: increasing $w^i(t)$ expands the cell by outwardly translating its boundaries, while decreasing it contracts the cell through inward boundary movement. Therefore, to adjust the cell size, the weight adjustment law (7) and the controller (8) are proposed. As in Line 2 of Algorithm 2, each

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(a) Only adjust the weight $w^i(t)$. (b) Only adjust the position $p^i(t)$.

Fig. 3. Effect of adjusting $w^i(t)$ and $p^i(t)$ on the cell $V^i(t)$.

aerial robot i updates its weight $w^i(t)$ by the weight adjustment law

$$\frac{\mathrm{d}}{\mathrm{d}t}w^i(t) = g_w \Delta s^i(t), \quad t \geq t_0, \tag{7}$$

where g_w is the positive gain and $\Delta s^i(t) = \bar{s}^i - s^i(t)$. This weight adjustment law is designed to make the cell mass $s^i(t)$ toward a target value $\bar{s}^i$ by shifting the boundaries. At the same time, in Line 2, each aerial robot i is driven by the controller

$$u^i(t) = g_2 \delta^i(t), \quad t \geq t_0, \tag{8}$$

where $\delta^i(t)$ is defined as

$$\delta^i(t) = \sum_{j\in\mathcal{N}^i} \left(\left(\Delta s^i - \Delta s^j\right) \int_{\partial V^{ij}} \phi_{\mathrm{L}}(q) \frac{q - p^i}{\|p^i - p^j\|} \mathrm{d}q \right).$$

Here ∂V^{ij} is the common edge of cells V^i and V^j. This controller integrates two distributed components to guide the motion of aerial robot i to further ensure the convergence of $s^i(t)$ to $\bar{s}^i$. The first is the influence of its position relative to all neighbors $\mathcal{N}^i$, quantified by terms like $(q - p^i)/\|p^i - p^j\|$. The second is the cell mass difference $(\Delta s^i - \Delta s^j)$. The weight adjustment law (7) and the controller (8) collaboratively ensure that the mass $s^i(t)$ of cell V^i converges to $\bar{s}^i$ to match its target load. This matching is crucial for making the most of the ground robots' coverage capabilities to enhance coverage performance.

Then we will analyze the convergence of the cell adjustment. First, Lemma 1 is introduced. By applying this lemma, Theorem 2 is proved, which guarantees the convergence of both the weight adjustment law (7) and the controller (8).

Lemma 1: Consider robot i and its neighbor $j \in \mathcal{N}^i$, for $\partial s^i/\partial w^i$ and $\partial s^i/\partial w^j$,

$$\frac{\partial s^i}{\partial w^i} = \sum_{j\in\mathcal{N}^i} \int_{\partial V^{ij}} \frac{\phi_{\mathrm{L}}(q)}{2\|p^j - p^i\|} \mathrm{d}q,$$

$$\frac{\partial s^i}{\partial w^j} = -\int_{\partial V^{ij}} \frac{\phi_{\mathrm{L}}(q)}{2\|p^i - p^j\|} \mathrm{d}q.$$

And for $\partial s^i/\partial p^i$ and $\partial s^j/\partial p^i$,

$$\frac{\partial s^i}{\partial p^i} = \sum_{j\in\mathcal{N}^i} \int_{\partial V^{ij}} \phi_{\mathrm{L}}(q) \frac{q - p^i}{\|p^i - p^j\|} \mathrm{d}q,$$

$$\frac{\partial s^j}{\partial p^i} = -\int_{\partial V^{ij}} \phi_{\mathrm{L}}(q) \frac{q - p^i}{\|p^i - p^j\|} \mathrm{d}q.$$

Proof: A detailed proof is derived in Appendix B. ∎

Theorem 2: When each aerial robot i executes weight adjustment law (7) and is driven by the controller (8), $s^i(t)$ converges to $\bar{s}^i$ for all $i \in A$ as $t \to \infty$.

Proof: Define the function $V(t) = \frac{1}{2}\sum_{i=1}^{K} \left(\Delta s^i(t)\right)^2$, then we can obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}V(t) = -\sum_{i=1}^{K} \left(\Delta s^i \left(\sum_{j=1}^{K} \frac{\partial s^i}{\partial w^j} \frac{\mathrm{d}w^j}{\mathrm{d}t} + \sum_{j=1}^{K} \frac{\partial s^i}{\partial p^j} \frac{\mathrm{d}p^j}{\mathrm{d}t} \right) \right).$$

We can equivalently write $\mathrm{d}V(t)/\mathrm{d}t = a + b$, where

$$a = -\sum_{i=1}^{K} \left(\Delta s^i \left(\sum_{j=1}^{K} \frac{\partial s^i}{\partial w^j} \frac{\mathrm{d}w^j}{\mathrm{d}t} \right) \right),$$

$$b = -\sum_{i=1}^{K} \left(\Delta s^i \left(\sum_{j=1}^{K} \frac{\partial s^i}{\partial p^j} \frac{\mathrm{d}p^j}{\mathrm{d}t} \right) \right).$$

According to Lemma 1, for a, we have

$$\begin{aligned} a &= -\sum_{i=1}^{K} \left(\Delta s^i \left(\frac{\partial s^i}{\partial w^i} \frac{\mathrm{d}w^i}{\mathrm{d}t} + \sum_{j\in\mathcal{N}^i} \frac{\partial s^i}{\partial w^j} \frac{\mathrm{d}w^j}{\mathrm{d}t} \right) \right) \\ &= -\frac{g_w}{2} \sum_{i=1}^{K} \left(\sum_{j\in\mathcal{N}^i} \left((\Delta s^i)^2 \int_{\partial V^{ij}} \frac{\phi_{\mathrm{L}}(q)}{\|p^j - p^i\|} \mathrm{d}q \right) \right. \\ &\quad \left. - \sum_{j\in\mathcal{N}^i} \left(\Delta s^i \Delta s^j \int_{\partial V^{ij}} \frac{\phi_{\mathrm{L}}(q)}{\|p^i - p^j\|} \mathrm{d}q \right) \right). \end{aligned}$$

Denote $g^{ij} = \int_{\partial V^{ij}} \phi_{\mathrm{L}}(q)/\|p^i - p^j\| \mathrm{d}q$, we have $g^{ij} = g^{ji}$. Rearranging the expression for a, we can further obtain

$$\begin{aligned} a &= -\frac{g_w}{2} \sum_{i=1}^{K} \sum_{j\in\mathcal{N}^i} \left(g^{ij} \left((\Delta s^i)^2 - \Delta s^i \Delta s^j \right) \right) \\ &= -\frac{g_w}{2} \left(\sum_{i=1}^{K} \sum_{j\in\mathcal{N}^i} \left(g^{ij} (\Delta s^i)^2 \right) - \sum_{i=1}^{K} \sum_{j\in\mathcal{N}^i} \left(g^{ij} \Delta s^i \Delta s^j \right) \right) \\ &= -\frac{g_w}{2} \left(\frac{1}{2} \sum_{i=1}^{K} \sum_{j\in\mathcal{N}^i} \left(g^{ij} \left((\Delta s^i)^2 + (\Delta s^j)^2 \right) \right) \right. \\ &\quad \left. - \frac{1}{2} \sum_{i=1}^{K} \sum_{j\in\mathcal{N}^i} \left(2 g^{ij} \Delta s^i \Delta s^j \right) \right) \\ &= -\frac{g_w}{4} \sum_{i=1}^{K} \sum_{j\in\mathcal{N}^i} g^{ij} \left(\Delta s^i - \Delta s^j \right)^2 \leq 0. \end{aligned}$$

According to Lemma 1, we have

$$\begin{aligned} \frac{\partial V(t)}{\partial p^i} &= \sum_{j=1}^{K} \left(\Delta s^j \frac{\partial \Delta s^j}{\partial p^i} \right) = -\sum_{j=1}^{K} \left(\Delta s^j \frac{\partial s^j}{\partial p^i} \right) \\ &= -\Delta s^i \sum_{j\in\mathcal{N}^i} \int_{\partial V^{ij}} \phi_{\mathrm{L}}(q) \frac{q - p^i}{\|p^i - p^j\|} \mathrm{d}q \\ &\quad - \sum_{j\in\mathcal{N}^i} \Delta s^j \int_{\partial V^{ij}} \phi_{\mathrm{L}}(q) \frac{p^i - q}{\|p^i - p^j\|} \mathrm{d}q \\ &= -\delta^i. \end{aligned}$$

Then, we can obtain

$$b = \sum_{i=1}^{K} \frac{\partial V(t)}{\partial p^i} \frac{\mathrm{d}p^i}{\mathrm{d}t} = -\sum_{i=1}^{K} \delta^i \frac{\mathrm{d}p^i}{\mathrm{d}t} = -g_2 \sum_{i=1}^{K} (\delta^i)^2 \leq 0.$$

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Therefore, we have $\mathrm{d}V(t)/\mathrm{d}t = a + b \leq 0$. When this is equal to zero, we have $a = b = 0$, at which point $\Delta s^i = \Delta s^j$ for all $i, j \in A$. Since $\sum_{i=1}^{K} \Delta s^i = 0$, it can be concluded that $\Delta s^i = 0, \forall i \in A$. According to LaSalle's invariance principle, the system converges to the largest invariant set contained within the set of all points where $\mathrm{d}V(t)/\mathrm{d}t = 0$, which implies that $s^i(t)$ converges to $\bar{s}^i$ for all $i \in A$ as $t \to \infty$. ■

V. Distributed Guidance and Coverage

This section introduces a distributed guidance and coverage scheme, along with an analysis of the computational complexity of Algorithm 2. While the aerial robots start to adjust the size of their cells from t_0, they simultaneously guide the ground robots to achieve high-resolution coverage. This enables the ground robots to overcome the limitations of their sensing range, driving them to achieve better coverage of a wider domain. As in Lines 2 to 2 of Algorithm 2, the scheme can be divided into three phases, which achieve distributed guidance of ground robots through the exchange and comparison of information between aerial robot i and its neighbors $\mathcal{N}^i$.

1. Offering Phase: First, aerial robot i calculates

$$x^i(t) = n^i(t) - \bar{n}^i, \tag{9}$$

which refers to the difference between the real-time number of ground robots $n^i(t)$ and the target number $\bar{n}^i$ within cell V^i. Then, aerial robot i obtains the minimum difference $\underline{x}^i(t)$ among neighbors, the set of aerial robots with this minimum difference $\underline{\mathcal{N}}^i(t)$, i.e.,

$$\begin{aligned} \underline{x}^i(t) &= \min_{j \in \mathcal{N}^i} x^j(t), \\ \underline{\mathcal{N}}^i(t) &= \left\{ j \in \mathcal{N}^i | x^j(t) = \underline{x}^i(t) \right\}, \end{aligned} \tag{10}$$

and chooses j_r randomly from $\underline{\mathcal{N}}^i(t)$. If $x^i(t) > x^{j_r}(t)$, it sends offer to aerial robot j_r.

2. Accepting Phase: Aerial robot i checks its received offer

$$M_1 = \left\{ \left(j_1, x^{j_1}(t)\right), \ldots, \left(j_m, x^{j_m}(t)\right) \right\}. \tag{11}$$

If $|M_1| > 0$, i.e., it has received offers, then it obtains the maximum difference $\overline{x}^i(t)$ among these aerial robots and the set of aerial robots with this maximum difference $\overline{\mathcal{N}}^i(t)$, i.e.,

$$\begin{aligned} \overline{x}^i(t) &= \max_{s \in \{1,\ldots,m\}} x^{j_s}(t), \\ \overline{\mathcal{N}}^i(t) &= \left\{ j_s \in M_1 | x^{j_s}(t) = \overline{x}^i(t) \right\}. \end{aligned} \tag{12}$$

Aerial robot i chooses h randomly from $\overline{\mathcal{N}}^i(t)$ and sends acceptance to aerial robot h.

3. Passing Phase: Aerial robot i checks its acceptance $M_2 = \{j\}$. If $|M_2| > 0$, i.e., it has received acceptance, then it uses onboard sensors (such as cameras) to detect ground robots in its cell, communicates with them to obtain their positions, identifies which robot is the closest to the target location $p^j(t)$, computes the control input (13) and transmits it via communication to guide the corresponding ground robot from V^i to V^j:

$$\begin{aligned} u_{r^*}(t) &= g_3 \left(p^j(t) - p_{r^*}(t) \right), \\ r^* &= \arg\min_r \| p^j(t) - p_r(t) \|, r \in \mathcal{J}^i(t), \end{aligned} \tag{13}$$

where g_3 is the positive control gain. If it has not received acceptance, it still uses its onboard sensors to detect ground robots in its cell, communicates with them, and directs them to execute the controller (14) to complete high-resolution coverage:

$$u_r(t) = g_4 \left(C_r^{\lim}(t) - p_r(t) \right), r \in \mathcal{J}^i(t), \tag{14}$$

where g_4 is the positive control gain, $C_r^{\lim}(t)$ is the mass centroid of $V_r^{\lim}(t)$ calculated by $\phi_{\mathrm{H}}(q)$. And $V_r^{\lim}(t)$ is the intersection region between ground robot r's cell $V_r(t)$ and its limited sensing region.

Then we will discuss the computational complexity of Algorithm 2. For Algorithm 2, each aerial robot i updates its weight using the weight adjustment law (7). This requires calculating s^i by integrating over its Voronoi cell V^i, costing $\mathcal{O}\left(M^i\right)$, where M^i is the number of discrete grid points for integration in V^i. The control input calculation (8) involves boundary integrals over ∂V^{ij}, costing $\mathcal{O}\left(d^i \cdot B^{ij}\right)$, where d^i is the number of aerial robot i's neighbors and B^{ij} is the number of discrete points on ∂V^{ij}. For each ground robot r, the coverage controller (14) computes the centroid within its limited sensing region, costing $\mathcal{O}(L_r)$, where L_r is the number of discrete grid points in that region. The guided controller (13) requires two communication processes: one with neighboring aerial robots to determine the target cell for guidance, costing $\mathcal{O}\left(d^i\right)$, and another with ground robots within the cell to coordinate their movement, costing $\mathcal{O}\left(n^i\right)$. Overall, the complexity is

$$\mathcal{O}\left(\sum_{i=1}^{K} M^i + \sum_{i=1}^{K} \sum_{j \in \mathcal{N}^i} d^i \cdot B^{ij} + \sum_{r=1}^{N} L_r + \sum_{i=1}^{K} \left(d^i + n^i\right) \right).$$

This is equal to

$$\mathcal{O}\left(M + \sum_{i=1}^{K} \sum_{j \in \mathcal{N}^i} d^i \cdot B^{ij} + \sum_{r=1}^{N} L_r + \sum_{i=1}^{K} d^i + N \right),$$

where M is the number of discrete grid points in the whole domain. Since $\sum_{i=1}^{K} \sum_{j \in \mathcal{N}^i} d^i \cdot B^{ij} \leq M$, $\sum_{r=1}^{N} L_r \leq M$, and the average number of Voronoi neighbors per aerial robot is bounded by a constant (at most six [40]), the computational complexity of Algorithm 2 can also be expressed as $\mathcal{O}(M + K + N)$.

Finally, we will analyze the convergence of the guidance and coverage scheme. What we need to prove is that $\lim_{t \to \infty} x^i(t) = 0, \forall i \in A$, which means that the number of ground robots $n^i(t)$ in each aerial robot i's cell converges to the target number $\bar{n}^i$. First, we need two lemmas: Lemma 2 and Lemma 3. Define

$$\begin{aligned} m(t) &= \min_{i \in A} x^i(t), \quad M(t) = \max_{i \in A} x^i(t), \\ D(t) &= M(t) - m(t). \end{aligned}$$

Lemma 2: In the execution of the guidance and coverage algorithm, for any time $t_1 \geq t_0$ such that $D(t_1) \geq 2$, there exists a finite time $t^* > t_1$ such that the probability $\mathrm{P}\left(\mathrm{D}(\mathrm{t}^*) < \mathrm{D}(\mathrm{t}_1)\right) > 0$.

Proof: A detailed proof is derived in Appendix C. ■

Define $\mathcal{X}(t) = [x^1(t), x^2(t), \ldots, x^K(t)]^\top$, then we introduce Lemma 3.

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Lemma 3 ([41, Theorem 2]): Consider a distributed algorithm, the following conditions are met:

(C1) for any initial $\mathcal{X}(t_0)$, at any time in the execution, the value of $\mathcal{X}(t)$ lies in a finite set $\mathscr{X}$,

(C2) for any state $\mathcal{X}(t) = \mathcal{X}$, there exists a finite time t_x such that $\mathrm{P}(\mathcal{X}(t + t_x) = \mathbf{0}_K \mid \mathcal{X}(t) = \mathcal{X}) > 0$,

(C3) if $\mathcal{X}(t) = \mathbf{0}_K$, then $\mathcal{X}(t') = \mathbf{0}_K$ for all $t' \geq t$.

In the execution of such an algorithm, for any initial $\mathcal{X}(t_0)$ such that $\sum_{i=1}^{K} x^i(t_0) = 0$, we have $\lim_{t\to\infty} \mathcal{X}(t) = \mathbf{0}_K$.

Then we use Lemmas 2 and 3 to prove Theorem 3.

Theorem 3: In the execution of the guidance and coverage algorithm, $\lim_{t\to\infty} \mathcal{X}(t) = \mathbf{0}_K$.

Proof: In the execution of the guidance and coverage algorithm, $m(t)$ is non-decreasing and $M(t)$ is non-increasing, which means that $D(t)$ is non-increasing. Therefore, in graph $\mathcal{G}$, at any time $t \geq t_0$, $x^i(t)$ is between $m(t_0)$ and $M(t_0)$. That is, there can be at most $D(t_0)+1$ different values for all $t \geq t_0$. As a result, the value of $\mathcal{X}(t)$ lies in a finite set $\mathscr{X}$, where a trivial upper bound on the size of $\mathscr{X}$ is $(D(t_0) + 1)^K < \infty$.

According to Lemma 2, there exists a finite time $\tilde{t} > t$ such that $\mathrm{P}\left(D(\tilde{t}) < 2\right) > 0$. Because $\sum_{i=1}^{K} x^i(t_0) = 0$, at time $\tilde{t}$, $\mathrm{P}\left(\mathcal{X}(\tilde{t}) = \mathbf{0}_K \mid \mathcal{X}(t) = \mathcal{X}\right) = \mathrm{P}\left(D(\tilde{t}) < 2\right) > 0$.

Since $m(t)$ is non-decreasing and $M(t)$ is non-increasing, we have that if $\mathcal{X}(t) = \mathbf{0}_K$, then for all $t' \geq t$,

$$M(t') \leq M(t) = 0, \quad m(t') \geq m(t) = 0,$$

which means that $\mathcal{X}(t') = \mathbf{0}_K$.

Therefore, the execution of guidance and coverage algorithm satisfies (C1), (C2), and (C3) in Lemma 3. According to Lemma 3, $\lim_{t\to\infty} \mathcal{X}(t) = \mathbf{0}_K$, which completes the proof. ■

VI. Simulations and Experiments

In this section, we validate the effectiveness of the proposed scheme through simulations and experiments, and compare its performance with three other schemes. More details of the experiments can be found in the supplemental video.

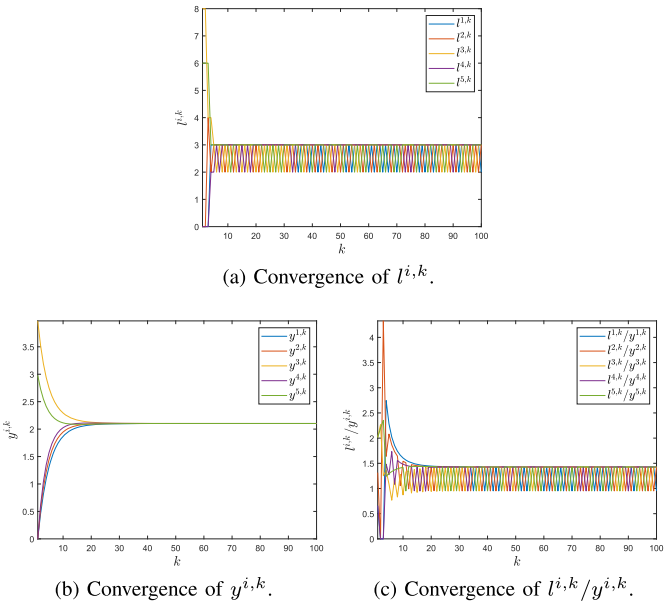
A. Simulation Results

In the simulation, the field of the ground domain D^G is chosen to be a 3×3 rectangle domain with

$$\phi(x,y) = 1.2\exp\left(-(x+1)^2-(y+1)^2\right)+1.2\exp\left(-(x+1)^2 - (y-1)^2\right) + \exp\left(-(x-1)^2 - y^2\right).$$

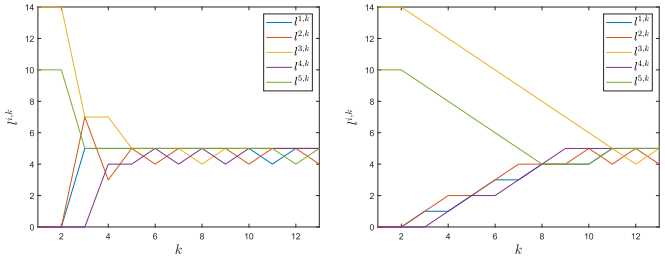
The scalar field is modeled as a weighted sum of exponential functions based on the Gaussian distribution observed from instantaneous point sources like fires or chemical leaks [42], [43]. We employ four robot teams of $N = 14$ ground robots each, assigned to different coverage schemes.

- Team 1 applies the scheme in [6] without any modifications. It assumes an unlimited sensing range for ground robots.
- Team 2 also applies the scheme in [6], but the ground robots' sensing range is limited to 0.25 length units.
- Team 3 applies the scheme in [29] and utilizes $K = 5$ aerial robots. The aerial robots are configured with a sensing range of 1.2 length units and a communication range of 1.8 length units, while the ground robots' sensing is limited to 0.25 length units.
- Team 4 utilizes the proposed scheme, also with $K = 5$ aerial robots. The aerial robots are configured with a sensing range of 1.2 length units and a communication range of 1.8 length units, while the ground robots' sensing is limited to 0.25 length units.



(a) Convergence of $l^{i,k}$. (b) Convergence of $y^{i,k}$. (c) Convergence of $l^{i,k}/y^{i,k}$.

Fig. 4. Visualizations of convergence for Algorithm 1.



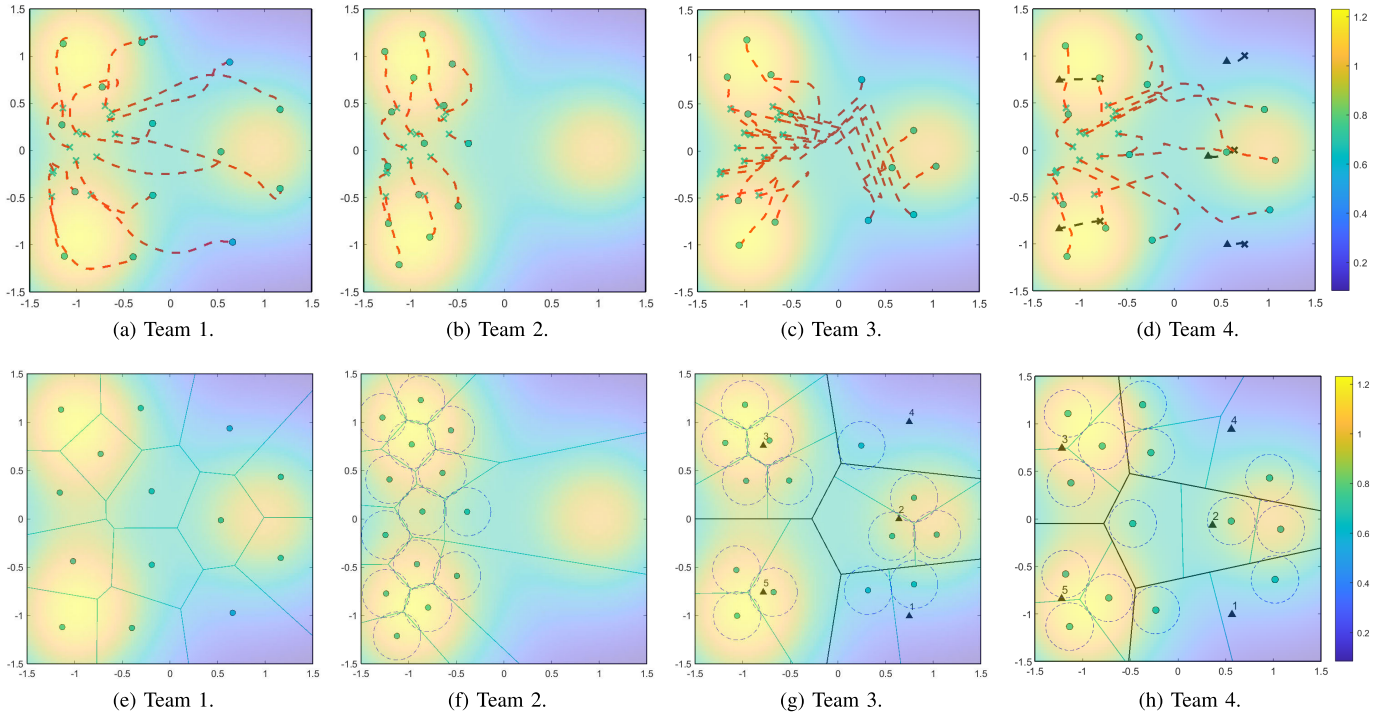
(a) With the original exchanged load. (b) With the fixed exchanged load 1.

Fig. 5. Convergence under two exchanged load values.

Teams 2, 3, and 4 operate under the constraint of limited sensing capabilities for the ground robots, with Team 2 serving as the baseline for this constrained scenario. In contrast, Team 1 represents the ideal case where ground robots have unlimited large sensing, achieving superior coverage performance. For these four teams, we maintain consistent initial positions for both ground and aerial robots. Additionally, g_1 is set to 0.4 for both Teams 3 and 4. For Team 4, parameter values are set to $g_w = 0.09, g_2 = 0.1, g_3 = 0.2, g_4 = 0.4$. The high-resolution means the ground robots can detect changes in $\phi(q)$ as small as 1×10^{-5}, while the low-resolution means the aerial robots can detect changes as small as 1×10^{-2}.

To validate the effectiveness of Algorithm 1, we set $k_{\text{total}} = 100$ and $g_y = 0.1$. During the execution of Algorithm 1, the value $l^{i,k}$, $y^{i,k}$ and $l^{i,k}/y^{i,k}$ evolves with k are shown in Figs. 10(a), 10(b) and 10(c) respectively. It can be observed that $l^{i,k}$ eventually vibrates between 3 and 2, which arises

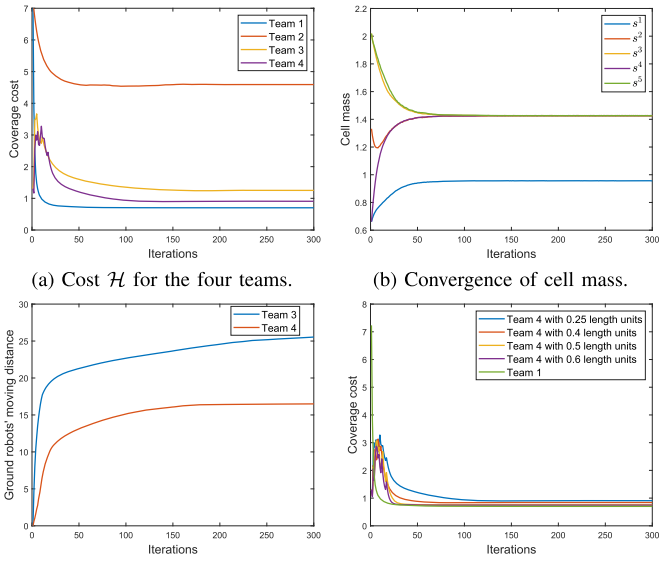
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(a) Team 1. (b) Team 2. (c) Team 3. (d) Team 4.

(e) Team 1. (f) Team 2. (g) Team 3. (h) Team 4.

Fig. 6. The trajectories of robots in simulations (a) to (d) and their final configurations in simulations (e) to (h). The initial positions of the ground robots are denoted with green X markers, the trajectories with red dotted lines, the final positions with solid green circles, the sensing regions with blue dashed circles, and the ground robots' boundaries with solid green lines. The initial positions of the aerial robots are denoted with black X markers, the trajectories with black dotted lines, the final positions with solid black triangles, and the aerial robots' boundaries with solid black lines.



(a) Cost $\mathcal{H}$ for the four teams. (b) Convergence of cell mass.

(c) Accumulative distance of all robots. (d) Cost $\mathcal{H}$ for different radius R.

Fig. 7. The coverage performance in the simulation.

because 14 ground robots cannot be evenly distributed among 5 aerial robots. Hence, we stop at $k_{\text{total}} = 100$ and obtain $\bar{n}^1 = 2, \bar{n}^i = 3, i = 2,3,4,5$. From Fig. 10(b), we can see that $y^{i,k}$ eventually converges to $N/S = 2.1024$. Consequently, as is shown in Fig. 10(c), $l^{1,k}/y^{1,k}$ converges to 0.9513, while $l^{i,k}/y^{i,k}, i = 2,3,4,5$ converge to 1.4269, resulting in $\bar{s}^1 = 0.9513$ and $\bar{s}^i = 1.4269, i = 2,3,4,5$. Then, we have $\bar{s}^i/\bar{n}^i = S/N = 0.4756, i = 1,2,3,4,5$, implying that the mass value of each cell matches its load value.

In Algorithm 1, we further modify Line 1 and Line 1 by updating the setting as $l^{i,k+1} \leftarrow l^{i,k} + 1$ and $l^{i,k+1} \leftarrow l^{i,k} - 1$ respectively. This change reduces the exchanged load value from $\lceil (l^{i,k} - l^{h,k})/2 \rceil$ to a fixed value of 1. We then examine how this modification affects the convergence rate. As shown in Fig. 5, under the same initial values, reducing the exchanged load value slows down the convergence: while the original algorithm converges at $k = 5$, the modified version requires $k = 11$ iterations to converge.

Figs. 6(a), 6(b), 6(c) and 6(d) demonstrate the trajectories of four teams, and Figs. 6(e), 6(f), 6(g) and 6(h) illustrate the final configurations: (a) and (e) correspond to Team 1, (b) and (f) correspond to Team 2, (c) and (g) correspond to Team 3, and (d) and (h) correspond to Team 4. The coverage cost function (5) is used to evaluate the coverage performance in Fig. 7(a). Teams 3 and 4 exhibit lower costs with fluctuations at earlier iterations. This is because the ground robots in these teams have not yet fully covered the entire domain D^G at earlier iterations, and the ground robots are still being guided towards cells with fewer ground robots, resulting in fluctuations in the coverage cost rather than a decline. As iteration goes on, the ground robots achieve complete coverage, making costs comparable.

In Fig. 7(a), the coverage cost of Team 2 is higher. This outcome can be attributed to the constraints imposed by the limited sensing, rendering it difficult for the ground robots to cover the right side. In contrast, Team 4 achieves a coverage cost that approximates that of Team 1 and it is lower than Team 3. This is because in the proposed scheme, each aerial robot adjusts its cell size to match the number of

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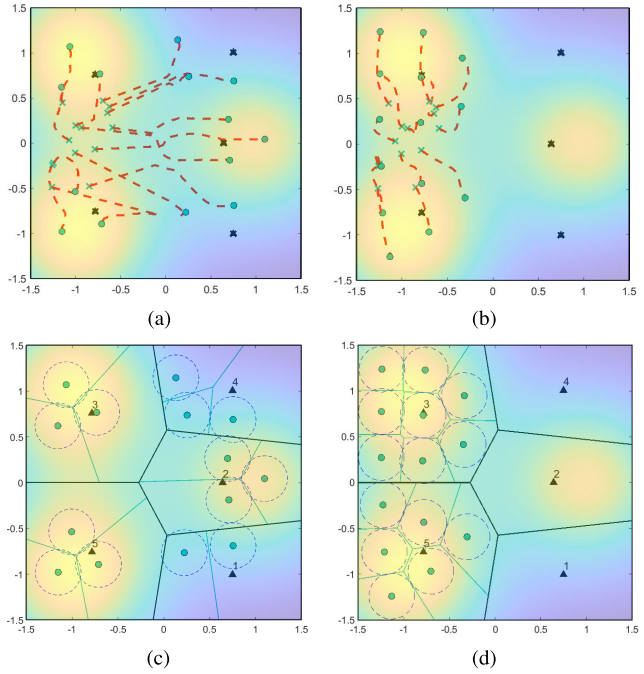


Fig. 8. Trajectories and final configurations of aerial and ground robots without cell adjustment and distributed guidance.

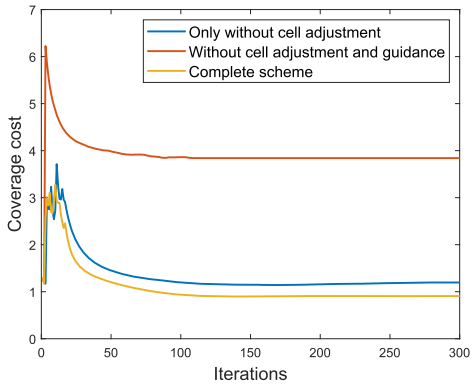


Fig. 9. Coverage cost under the ablation study.

ground robots within its cell, thereby fully leveraging each ground robot's coverage capability to achieve better coverage. Fig. 7(b) demonstrates the convergence of cell mass in Team 4. Under the combined effect of the weight adjustment law (7) and the controller (8), the aerial robots can quickly adjust the size of their cells, enabling the cell mass to converge to the target values. Fig. 7(c) illustrates the evolution of the accumulated travel distance of all ground robots in Teams 3 and 4. The accumulated travel distance of all ground robots in Team 4 is less than that of Team 3. This is attributed to the guide controller (13), which first moves the ground robot closest to its destination; in contrast, all ground robots within a cell in Team 3 are moved simultaneously. Fig. 7(d) illustrates the coverage cost of Team 4 for different sensing radii. As the radius increases, the coverage cost gets closer to that of Team 1.

1) Ablation Study: An ablation study is conducted to evaluate the effectiveness of each proposed module.

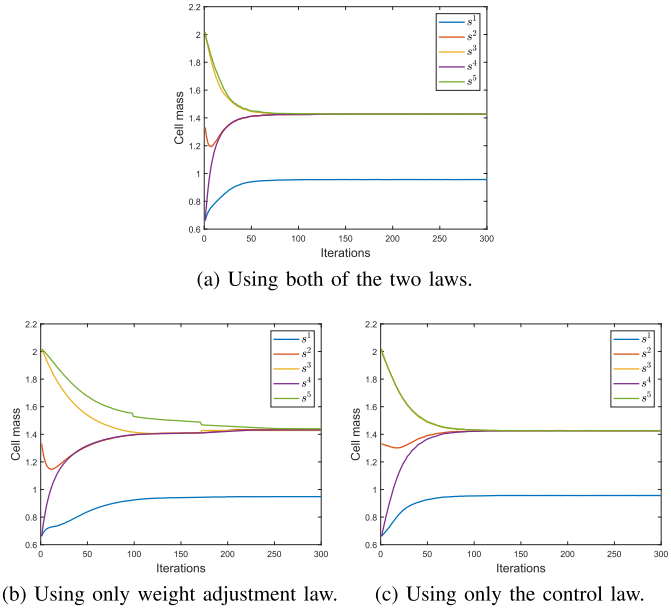


Fig. 10. Convergence of cell mass under three configurations.

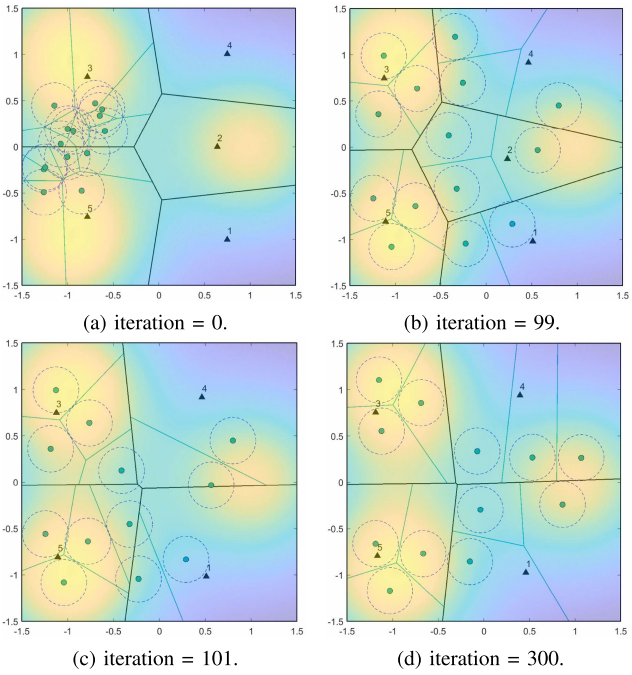
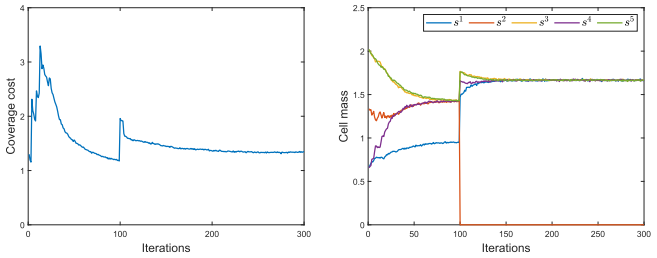


Fig. 11. Evolution of aerial and ground robot configurations.

Fig. 8(a) and 8(c) illustrate the trajectories and final configurations of the aerial and ground robots when the cell adjustment is disabled, while Fig. 8(b) and 8(d) depict the results without both the cell adjustment and the distributed guidance. Without cell adjustment, the number of ground robots in each aerial robot's cell does not match the corresponding cell mass, which fails to fully leverage the coverage capability of each ground robot. The absence of distributed guidance leads to poor coverage performance: the ground robots only cover the left domain, leaving a large portion of the right domain uncovered. A comparison of the coverage cost under three configurations is provided in Fig. 9: the complete scheme,

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(a) Evolution of coverage cost. (b) Convergence of cell mass.

Fig. 12. Evolution of coverage cost and cell mass.

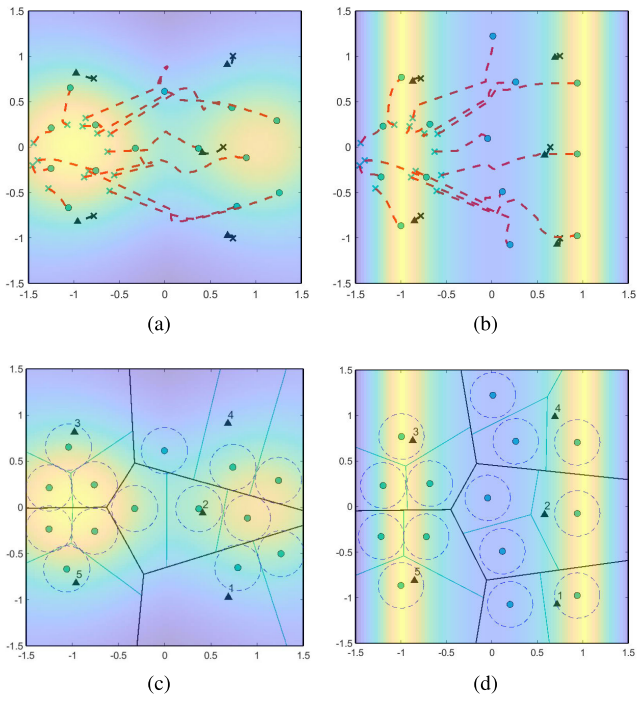
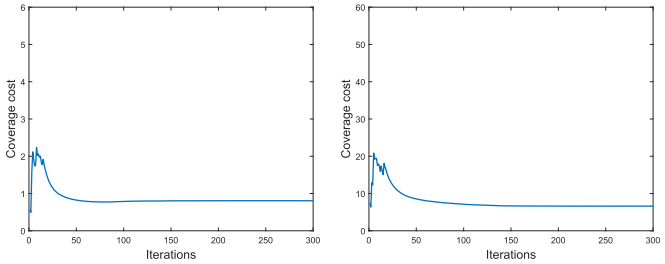
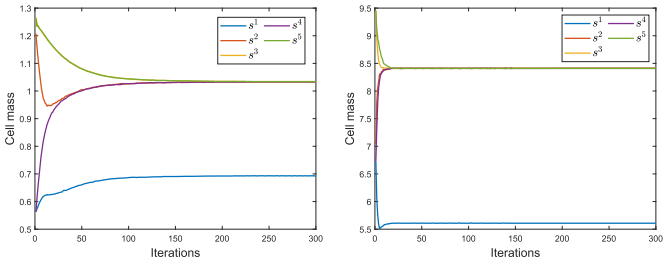


Fig. 13. Trajectories and final configurations of aerial and ground robots under other scalar fields.



(a) Under the scalar field $\phi_1(x, y)$. (b) Under the scalar field $\phi_2(x, y)$.

Fig. 14. Evolution of coverage cost under other two scalar fields.



(a) Under the scalar field $\phi_1(x, y)$. (b) Under the scalar field $\phi_2(x, y)$.

Fig. 15. Convergence of cell mass under other two scalar fields.

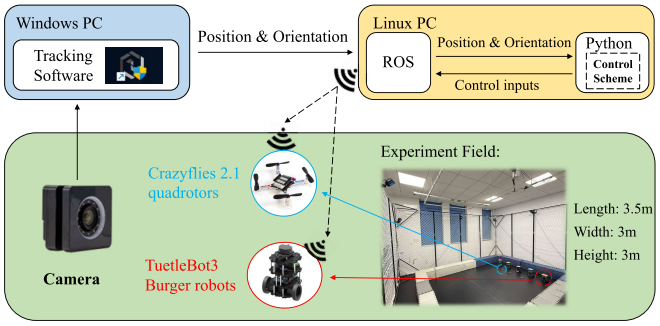


Fig. 16. Experimental system schematic.

the scheme without cell adjustment only, and the scheme without both cell adjustment and distributed guidance. The results confirm that both modules contribute to improving the coverage performance. Moreover, Fig. 10 compares the cell mass convergence under three configurations: using both laws simultaneously, using only the weight adjustment law (7), and using only the control law (8). The results demonstrate that the combined utilization of both laws leads to a faster convergence rate of the cell mass compared to using either law alone.

2) Robustness: To evaluate the robustness of the proposed scheme, we conduct the test under asynchronous execution with the presence of sensing noise and robot failures during operation. The sensing noise is modeled as a zero-mean Gaussian white noise, whose standard deviation σ is set to 10 % of the peak amplitude of the field value $\phi(q)$, i.e., $\sigma = 0.1 \times \max(\phi(q))$. Fig. 11 illustrates the evolution of the robot configurations. At iteration 100, aerial robot 2 and two ground robots in the cell of aerial robot 4 experience a failure. This leads to a reduction in the number of ground robots to 12 and the number of aerial robots to 4. After that, the remaining four aerial robots re-balance the number of ground robots within their cells. Each aerial robot is now responsible for supervising 12/4 = 3 ground robots. Concurrently, the remaining four aerial robots re-adjust the size of their cells so that the cell mass matches the number of ground robots. Fig. 12 shows the evolution of the cell mass and the coverage cost. After iteration 100, cell mass s^2 decreases to zero, while the other cell masses re-converge to a balanced value with minor sensing fluctuations. A slight increase in the cost at iteration 100 is due to the failure of robots. As the remaining robots continue to perform the coverage, the cost continues to decrease and finally stabilizes.

We also test the proposed scheme under two non-differentiable field functions

$$\phi_1(x, y) = \exp\left(-(x+1)^2 - y^2\right) + \exp\left(-(x-1)^2 - y^2\right) + 0.1|x|,$$
$$\phi_2(x, y) = -0.1|x^2 - 1| + 5.$$

Fig. 13 illustrates the trajectories and final configurations of the robots operating in these two different scalar

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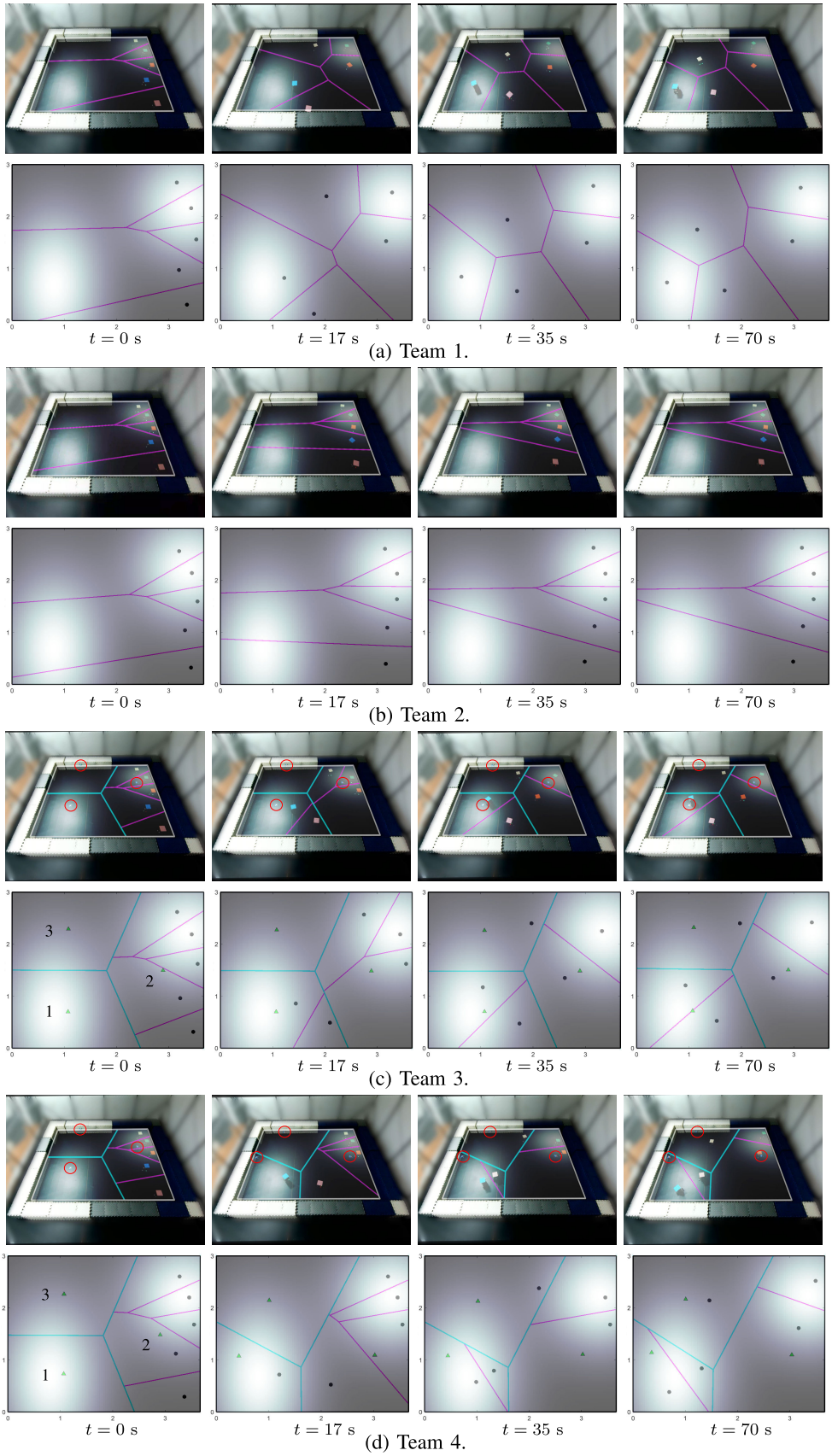
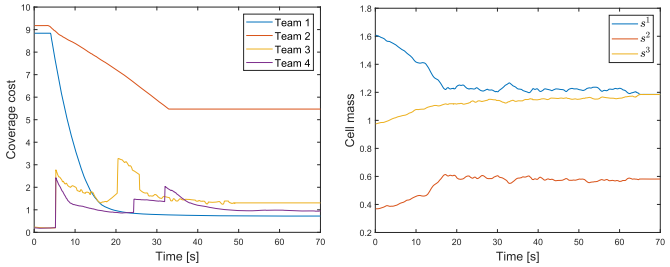


Fig. 17. Four teams achieve coverage. Purple lines denote the robots' cell boundaries, cyan lines denote the quadrotors' boundaries. Red circles highlight each quadrotor (oblique views), black circles mark Burger robots with green triangles denoting quadrotors (top views).

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(a) Cost $\mathcal{H}$ for the four teams. (b) Convergence of cell mass.

Fig. 18. The coverage performance in the experiment.

fields, (a) and (c): under scalar field $\phi_1(x, y)$, (b) and (d): under scalar field $\phi_2(x, y)$. The corresponding evolution of the coverage cost under these two scalar fields is presented in Fig. 14, while the convergence of the cell mass is shown in Fig. 15. The results demonstrate that the proposed scheme remains effective and performs well across both tested scalar fields.

Remark 5: Given that the proposed scheme is based on Voronoi coverage control, its stability analysis under asynchronous execution is analogous to the analysis provided in Proposition 3.3 of Section III.B in [6].

B. Experimental Results

In the experiment, we implement the proposed scheme using multiple Crazyflies 2.1 quadrotors [44] and TurtleBot3-Burger robots. The composition of the experimental testbed shown in Fig. 16 comprises two desktop computers, a FZMotion motion capture system, Crazyflies 2.1 quadrotors, and TurtleBot3-Burger robots. The motion capture system, utilizing eight cameras coupled with tracking software on a Windows PC, acquires positions and orientations at a rate of 10 ms. This data is then sent to a Linux PC running Ubuntu 20.04. In the Linux PC, the control inputs are computed based on the acquired data. These inputs are then transmitted to the Crazyflie via the Crazyradio and to the TurtleBot3-Burger via WiFi.

For Crazyflie 2.1 quadrotors, control is executed via the built-in PID controller, requiring only velocity inputs while ignoring complex motor control. We set $g_1 = 0.1$, $g_2 = 0.05$ and $g_w = 0.02$. The proposed control scheme, which is based on single integrator dynamics (4), is not applicable to the differential-drive configuration of TurtleBot3-Burger robots. Therefore, the control inputs need to be converted for Burger robots. The position and orientation of Burger robot j are denoted by $p_j = [x_j, y_j]^\top$ and θ_j, so the differential-drive configuration can be modeled by unicycle dynamics

$$\dot{x}_j = v_j \cos\theta_j, \quad \dot{y}_j = v_j \sin\theta_j, \quad \dot{\theta}_j = \omega_j,$$

where v_j and ω_j represent the linear velocity and angular velocity. Similar to [7], we can approximate the conversion of the single integrator dynamics (4) into unicycle dynamics as follows

$$v_j = g_5[\cos\theta_j, \sin\theta_j]u_j,$$
$$\omega_j = g_6 \arctan\left(\frac{[-\sin\theta_j, \cos\theta_j]u_j}{[\cos\theta_j, \sin\theta_j]u_j}\right),$$

where g_5 and g_6 are positive gains. And we set $g_5 = 0.2$ and $g_6 = 0.4$ in the experiment.

In the experiment, we also utilize four teams of robots to validate the four schemes previously tested in the simulation. Each team is tested using $N = 5$ Burger robots, and for Teams 3 and 4, $K = 3$ quadrotors are used. As shown in Fig. 16, the experimental domain is set as a 3.5 m $\times$ 3 m rectangular domain with scalar field

$$\phi(x, y) = \exp\left(-\frac{5(x-0.7)^2}{4} - \frac{(y-0.6)^2}{2}\right) + \exp\left(-(x-3)^2 - (y-2)^2\right).$$

For Teams 2-4, each Burger robot is set to have access to $\phi_{\mathrm{H}}(q)$ within a circular region of radius 0.4 m. For Teams 3-4, the sensing range of the aerial robots is set as 2 m, and the communication range is set as 2.4 m. The high-resolution means that the Burger robots can detect changes in $\phi(q)$ as small as 1×10^{-4}, while the low-resolution means the quadrotors can detect changes as small as 1×10^{-1}, as shown in Fig. 2. For the proposed scheme, each quadrotor obtains $\bar{n}^i$ and $\bar{s}^i$ through Algorithm 1: $\bar{n}^1 = 2, \bar{n}^2 = 1, \bar{n}^3 = 2$ and $\bar{s}^1 = 1.16, \bar{s}^2 = 0.58, \bar{s}^3 = 1.16$, achieving $\bar{s}^i/\bar{n}^i = 0.58$ for $i = 1, 2, 3$.

Figs. 17(a), 17(b), 17(c) and 17(d) illustrate experimental snapshots of four teams at different time instances. Each time instance includes both an oblique-view figure of the experiment and a corresponding top-view figure generated in MATLAB based on motion-capture data. In Fig. 17(d), at $t = 17$ s, a Burger robot is driven from quadrotor 2's cell (which contains multiple Burger robots) to quadrotor 1's cell (which contains no Burger robots previously). At $t = 35$ s, the number of Burger robots in each quadrotor's cell reaches the target number, then Burger robots start to achieve high-resolution coverage. By $t = 70$ s, complete coverage is achieved.

Fig. 18(a) illustrates the evolution of coverage cost for four teams in the experiment. It is observed that Team 4 achieves the lowest coverage cost among the teams with sensing-limited Burger robots, and its cost approximates that of Team 1, which operates with unlimited large sensing. Fig. 18(b) demonstrates the convergence of cell masses for three quadrotors under the proposed scheme, showing that $s^i(t)$ converges to $\bar{s}^i$. According to Fig. 17(d), the number of Burger robots in each quadrotor's cell $n^i(t)$ finally reaches its target value $\bar{n}^i$. The resulting ratios $s^i/n^i = 0.58, i = 1, 2, 3$ indicate that the proposed scheme makes the most of the coverage capability of each Burger robot to achieve better coverage.

VII. Conclusion

In this paper, we propose a fully distributed coverage control scheme for air-ground robot systems with heterogeneous sensing capabilities. Initially, aerial robots quickly cover the entire domain, perceiving coarse but broader environmental information. Then, aerial robots cooperatively guide the ground robots to overcome the limitations of their short sensing range, enabling them to achieve better coverage. Convergence is proved and effectiveness validated through experiments. Future

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work will address coverage limitations due to obstacles in a domain, aiming to extend this scheme to non-convex environments. And we plan to explore its application in manifold coverage.

APPENDIX A
THE PROOF OF THEOREM 1

Proof: Due to space constraints and for the sake of completeness, we present a concise proof here, which is similar in spirit to the proofs of [39, Claim 5.8] and [45, Theorem 4.1].

Define $l^k_{\min} = \min_{i\in A} l^{i,k}$, the set $L^k_{\min} = \{i \in A | l^{i,k} = l^k_{\min}\}$, and the function $F(k) = (K+1)\left(\lfloor N/K\rfloor - l^k_{\min}\right) + \left|L^k_{\min}\right|$, where $\left|L^k_{\min}\right|$ represents the cardinality of $L^k_{\min}$.

If there exists a node $i \in L^k_{\min}$ which has a neighbor with the load exceeding $l^k_{\min}+2$, then at $k+1$, $F(k+1) < F(k)$. If all neighbors of nodes in $L^k_{\min}$ have load $l^k_{\min}+1$, the random selection of a neighbor by each node induces a stochastic process. In this scenario, the positions of the minimum-load nodes transfer in $\mathcal{G}$, and this state transference can be modeled as a homogeneous Markov chain. According to the properties of a homogeneous Markov chain, during the execution of Algorithm 1, when the convergence is not achieved, i.e., there exists some i whose $l^{i,k} \neq \lceil N/K\rceil$ or $\lfloor N/K\rfloor$ at time k, two cases hold:

- Case 1: When $\left|L^k_{\min}\right| \geq 2$, there exists a finite k_1 such that $\left|L^{k+k_1}_{\min}\right| < \left|L^k_{\min}\right|$ and $l^{k+k_1}_{\min} = l^k_{\min}$.
- Case 2: When $\left|L^k_{\min}\right| = 1$, there exists a finite k_1 such that $\left|L^{k+k_1}_{\min}\right| \geq \left|L^k_{\min}\right|$ and $l^{k+k_1}_{\min} \geq l^k_{\min}+1$.

For these two cases, we can all obtain $F(k+k_1) < F(k)$. Therefore, based on the above analysis, as $k\to\infty$, $F(k)$ will decrease until it reaches a minimum. At this point, $l^k_{\min}$ reaches $\lfloor N/K\rfloor$ and $\left|L^k_{\min}\right|$ is minimized, implying that $l^{i,k}$ converges to either $\lceil N/K\rceil$ or $\lfloor N/K\rfloor$.

Line 1 in Algorithm 1 can be expressed in matrix form as

$$Y^k = P_w Y^{k-1}, \quad P_w = I - g_y L_w, \quad L_w = W^{-1}L \tag{15}$$

where $Y^k = [y^{1,k}, y^{2,k}, \ldots, y^{K,k}]^\top$ is the column matrix, $W = \operatorname{diag}\left(s^{1,0}, \ldots, s^{K,0}\right)$ is the weighted matrix, $L = D - A$ is the Laplacian matrix of $\mathcal{G}$, D is a diagonal matrix with the out-degrees and A is the adjacency matrix of $\mathcal{G}$. Since $g_y < \min_{i\in A}\left(s^{i,0}/|\mathcal{N}^i|\right)$, we can obtain $P_{w,ii} = 1 - g_y|\mathcal{N}^i|/s^{i,0} > 0$ and $P_{w,ij} = g_y/s^{i,0} > 0$, which implies that P_w is irreducible nonnegative. According to the Perron-Frobenius theorem, we can obtain $\lim_{k\to\infty} P^k_w = e\frac{v^\top}{\|v\|}$, where e is the vector of all ones, and v is a positive vector satisfying that $P^\top_w v = v$.

From $P^\top_w v = v$, we have $L^\top_w v = 0$, hence $L^\top W^{-1} v = 0$. Since L is symmetric, $LW^{-1}v = 0$, implying $W^{-1}v = \beta e, \beta \in \mathbb{R}$ and thus $v = \beta We$. If $k\to\infty$, we have

$$\begin{aligned} Y^k &\overset{(15)}{=\!=} P^k_w Y^0 = e\frac{v^\top}{\|v\|}Y^0 = e\frac{\beta e^\top W^\top}{\|\beta We\|}Y^0 = \frac{W^\top}{\|W\|}Y^0 \\ &= \frac{\sum_{i\in A} s^{i,0}y^{i,0}}{\sum_{i\in A} s^{i,0}}e = \frac{\sum_{i\in A} n^{i,0}}{\sum_{i\in A} s^{i,0}}e = \frac{N}{S}e, \end{aligned}$$

which implies that $y^{i,k}$ converges to N/S. Therefore, $\bar{n}^i/\bar{s}^i$ is equal to N/S. ∎

APPENDIX B
THE PROOF OF LEMMA 1

Proof: Denote n^{ij} is the unit normal vector from V^i to V^j, for $\partial s^i/\partial w^i$ and $\partial s^i/\partial w^j$, we can obtain

$$\begin{aligned} \frac{\partial s^i}{\partial w^i} &= \frac{\partial}{\partial w^i}\int_{V^i}\phi_{\mathrm{L}}(q)\,\mathrm{d}q = \sum_{j\in\mathcal{N}^i}\int_{\partial V^{ij}}\phi_{\mathrm{L}}(q)\frac{\partial q^\top}{\partial w^i}n^{ij}\,\mathrm{d}q, \\ \frac{\partial s^i}{\partial w^j} &= \frac{\partial}{\partial w^j}\int_{V^i}\phi_{\mathrm{L}}(q)\,\mathrm{d}q = \int_{\partial V^{ij}}\phi_{\mathrm{L}}(q)\frac{\partial q^\top}{\partial w^j}n^{ij}\,\mathrm{d}q. \end{aligned}$$

For $q \in \partial V^{ij}$, it satisfies

$$\|p^i - q\|^2 - w^i = \|p^j - q\|^2 - w^j. \tag{16}$$

Taking the partial derivative of both sides of Eq. (16) with respect to w^i, we obtain

$$-2\frac{\partial q^\top}{\partial w^i}(p^i - q) - 1 = -2\frac{\partial q^\top}{\partial w^i}(p^j - q).$$

Rearranging the terms on both sides of the equation, we obtain

$$\frac{\partial q^\top}{\partial w^i}n^{ij} = \frac{\partial q^\top}{\partial w^i}\frac{(p^j - p^i)}{\|p^j - p^i\|} = \frac{1}{2\|p^j - p^i\|}. \tag{17}$$

Similarly, we can also obtain

$$\frac{\partial q^\top}{\partial w^j}n^{ij} = -\frac{1}{2\|p^i - p^j\|}. \tag{18}$$

Then, by substituting Eqs. (17) and (18) into the expressions for $\partial s^i/\partial w^i$ and $\partial s^i/\partial w^j$ respectively, we obtain

$$\begin{aligned} \frac{\partial s^i}{\partial w^i} &= \sum_{j\in\mathcal{N}^i}\int_{\partial V^{ij}}\frac{\phi_{\mathrm{L}}(q)}{2\|p^j - p^i\|}\,\mathrm{d}q, \\ \frac{\partial s^i}{\partial w^j} &= -\int_{\partial V^{ij}}\frac{\phi_{\mathrm{L}}(q)}{2\|p^i - p^j\|}\,\mathrm{d}q. \end{aligned}$$

For $\partial s^i/\partial p^i$ and $\partial s^j/\partial p^i$, we can obtain

$$\begin{aligned} \frac{\partial s^i}{\partial p^i} &= \sum_{j\in\mathcal{N}^i}\int_{\partial V^{ij}}\phi_{\mathrm{L}}(q)\frac{\partial q^\top}{\partial p^i}n^{ij}\,\mathrm{d}q, \\ \frac{\partial s^j}{\partial p^i} &= \int_{\partial V^{ij}}\phi_{\mathrm{L}}(q)\frac{\partial q^\top}{\partial p^i}n^{ji}\,\mathrm{d}q. \end{aligned}$$

Taking the partial derivative of both sides of Eq. (16) with respect to p^i and rearranging the terms, we obtain

$$\frac{\partial q^\top}{\partial p^i}n^{ij} = \frac{q - p^i}{\|p^i - p^j\|}.$$

Substituting the above expression into the expressions for $\partial s^i/\partial p^i$ and $\partial s^j/\partial p^i$ respectively, we obtain

$$\begin{aligned} \frac{\partial s^i}{\partial p^i} &= \sum_{j\in\mathcal{N}^i}\int_{\partial V^{ij}}\phi_{\mathrm{L}}(q)\frac{q - p^i}{\|p^i - p^j\|}\,\mathrm{d}q, \\ \frac{\partial s^j}{\partial p^i} &= -\int_{\partial V^{ij}}\phi_{\mathrm{L}}(q)\frac{q - p^i}{\|p^i - p^j\|}\,\mathrm{d}q, \end{aligned}$$

which completes the proof. ∎

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APPENDIX C
THE PROOF OF LEMMA 2

Proof: Since the maximum value $M(t)$ is non-increasing and the minimum value $m(t)$ is non-decreasing, to prove that there exists a finite time $t^* > t_1$ such that $\mathrm{P}(\mathrm{D}(\mathrm{t}^*) < \mathrm{D}(\mathrm{t}_1)) > 0$, it is equivalent to prove that: $\mathrm{P}(\mathrm{M}(\mathrm{t}^*) < \mathrm{M}(\mathrm{t}_1)) > 0$ or $\mathrm{P}(\mathrm{m}(\mathrm{t}^*) > \mathrm{m}(\mathrm{t}_1)) > 0$.

For any time t_1 such that $D(t_1) \geq 2$, we can always define two nonempty sets:

$$\mathscr{Q}(t_1) = \{i \in A | x^i(t_1) \leq M(t_1) - 2\},$$
$$\mathscr{M}(t_1) = \{i \in A | x^i(t_1) = M(t_1)\}.$$

And there may exist $\mathscr{O}(t_1) = \{i \in A | x^i(t_1) = M(t_1) - 1\}$, which may be empty. In the execution of the guidance and coverage algorithm, there exist two cases:

- **Case 1**: $|\mathscr{M}(t_1)| \leq |\mathscr{Q}(t_1)|$.
- **Case 2**: $|\mathscr{M}(t_1)| > |\mathscr{Q}(t_1)|$.

Case 1: Becasue $|\mathscr{M}(t_1)| \leq |\mathscr{Q}(t_1)|$, we will prove that $\mathrm{P}(\mathrm{M}(\mathrm{t}^*) < \mathrm{M}(\mathrm{t}_1)) > 0$, i.e., there exist $|\mathscr{M}(t_1)|$ time instants τ_i where $t_1 < \tau_1 \leq \tau_2 \leq \ldots \leq \tau_i \leq \ldots \leq \tau_{|\mathscr{M}(t_1)|} = t^*$, such that

$$\begin{aligned}
|\mathscr{M}(\tau_1)| &= |\mathscr{M}(t_1)| - 1, \\
|\mathscr{M}(\tau_2)| &= |\mathscr{M}(\tau_1)| - 1 = |\mathscr{M}(t_1)| - 2, \\
|\mathscr{M}(\tau_3)| &= |\mathscr{M}(\tau_2)| - 1 = |\mathscr{M}(t_1)| - 3, \\
&\vdots \\
|\mathscr{M}(\tau_{|\mathscr{M}(t_1)|-1})| &= |\mathscr{M}(\tau_{|\mathscr{M}(t_1)|-2})| - 1 \\
&= |\mathscr{M}(t_1)| - (|\mathscr{M}(t_1)| - 1) = 1.
\end{aligned}$$

And at $\tau_{|\mathscr{M}(t_1)|}$, that is, t^*, $M(t^{*+}) = M(t) - 1$ occurs. Because there is a jump in the value of $M(t)$ at t^*, a superscript + is used here to emphasize the state after the jump.

For some time τ_i, it is equivalent to prove that there is a positive probability that

- Case 1a: A node in $\mathscr{M}(\tau_i)$ provides an accepted offer to a node in $\mathscr{Q}(\tau_i)$ directly.
- Case 1b: A node in $\mathscr{M}(\tau_i)$ provides an accepted offer to a node i_O in $\mathscr{O}(\tau_i)$ and at the same time i_O provides an accepted offer to a node in $\mathscr{Q}(\tau_i)$.

For $t \in [\tau_{i-1}, \tau_i)$, it is possible that a node $i_M \in \mathscr{M}(t)$ provides an accepted offer to a node $i_O \in \mathscr{O}(t)$, however, i_O does not provide an accepted offer to a node in $\mathscr{Q}(t)$ at the same time. Under this situation, we can see i_O becomes a node in $\mathscr{M}(t^+)$ while i_M becomes a node in $\mathscr{O}(t^+)$. In this process, such i_M and i_O are randomly paired. Therefore, as the process continues, at some time τ_i, there exists a positive probability that a node from $\mathscr{M}(\tau_i)$ become a neighbor of a node from $\mathscr{O}(\tau_i)$ which is a neighbor of a node in $\mathscr{Q}(\tau_i)$, then again, the probability of Case 1b becomes positive.

According to the above analysis, in Case 1, we have $\mathrm{P}(\mathrm{M}(\mathrm{t}^*) < \mathrm{M}(\mathrm{t}_1)) > 0$.

Case 2: Similar to the analysis of Case 1, we define

$$\mathscr{R}(t_1) = \{i \in A | x^i(t_1) \geq m(t_1) + 2\},$$
$$\mathscr{N}(t_1) = \{i \in A | x^i(t_1) = m(t_1)\}.$$

Given that $|\mathscr{M}(t_1)| > |\mathscr{Q}(t_1)|$ and $\mathscr{M}(t_1) \subseteq \mathscr{R}(t_1)$, it follows that $|\mathscr{R}(t_1)| \geq |\mathscr{M}(t_1)| > |\mathscr{Q}(t_1)| \geq |\mathscr{N}(t_1)|$. Similar to the proof of Case 1, we can also prove that $\mathrm{P}(\mathrm{m}(\mathrm{t}^*) > \mathrm{m}(\mathrm{t}_1)) > 0$, so the detail is omitted here.

In conclusion, if $D(t_1) \geq 2$, there exists a finite time $t^* > t_1$ such that $\mathrm{P}(\mathrm{M}(\mathrm{t}^*) < \mathrm{M}(\mathrm{t}_1)) > 0$ or $\mathrm{P}(\mathrm{m}(\mathrm{t}^*) > \mathrm{m}(\mathrm{t}_1)) > 0$. Therefore, $\mathrm{P}(\mathrm{D}(\mathrm{t}^*) < \mathrm{D}(\mathrm{t}_1)) > 0$. ■

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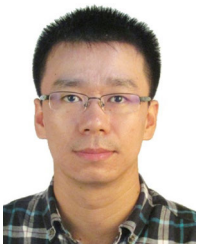
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